

Emergent Geometry from Entanglement: A Numerical Demonstration Using Heisenberg Spin Chains

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(Dated: June 21, 2026)

We present a fully numerical, model-independent demonstration that smooth spatial geometry can emerge directly from the entanglement structure of a pure quantum state, without any a priori geometric assumptions. Using exact diagonalization of the isotropic antiferromagnetic Heisenberg spin chain, we compute the ground-state mutual-information matrix and construct its associated graph Laplacian. A comprehensive suite of spectral and statistical diagnostics—including AICc model selection, extended finite-size scaling, bootstrap analysis, boundary-slope metrics, synthetic Laplacian benchmarking, and mutual-information decay exponents—shows that the information-derived operator exhibits the defining signatures of a one-dimensional Laplace–Beltrami operator with Neumann boundary conditions.

Our strongest quantitative result is a robust match between the measured mutual-information decay exponent and the analytic Luttinger-liquid prediction across the XXZ critical line. The lowest Laplacian eigenvalue λ_1 exhibits a statistically significant finite-size intercept under open boundary conditions (OBC), and this intercept becomes larger under periodic boundary conditions (PBC). This demonstrates that the mass scale is a genuine bulk property of the entanglement geometry rather than a boundary artefact. These results provide a concrete numerical pipeline for validating information-theoretic approaches to quantum gravity.

I. INTRODUCTION

A central idea in modern quantum gravity is that spacetime geometry is not fundamental, but emerges from structured patterns of quantum entanglement. This viewpoint appears in holographic dualities, tensor-network constructions, and quantum error-correcting codes, yet explicit numerical demonstrations in microscopic many-body systems remain scarce.

Here we treat the ground state of a simple quantum many-body system—the isotropic antiferromagnetic Heisenberg spin chain—as a raw relational dataset. From its mutual-information structure alone, we construct a graph Laplacian and analyze its spectral properties. We ask whether this operator behaves like a continuum Laplace–Beltrami operator on a one-dimensional manifold with Neumann boundary conditions, and how this behaviour changes between critical and gapped phases.

II. METHODS

A. Hamiltonian and ground state

We consider the open XXZ chain

$$H = J \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \quad (1)$$

with $J > 0$ and anisotropy Δ . Exact diagonalization yields the ground state $|\psi_0\rangle$ for system sizes up to $N = 18$ in the critical regime ($-1 < \Delta \leq 1$) and for selected gapped points $\Delta > 1$.

B. Mutual-information matrix

For each site i we define the reduced density matrix $\rho_i = \text{Tr}_{\bar{i}} |\psi_0\rangle\langle\psi_0|$, and for each pair (i, j) we define $\rho_{ij} = \text{Tr}_{\bar{i}\bar{j}} |\psi_0\rangle\langle\psi_0|$. The von Neumann entropies give the mutual information

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}). \quad (2)$$

C. Graph Laplacian construction

We define a normalized adjacency matrix

$$A_{ij} = \frac{I(i, j)}{\max_{k, \ell} I(k, \ell)}, \quad A_{ii} = 0, \quad (3)$$

and the associated graph Laplacian $L = D - A$ with $D_{ii} = \sum_j A_{ij}$.

III. RESULTS

A. Zero mode and Neumann boundary conditions

The lowest eigenvector of L is constant to machine precision, with $\text{std}(v_0)/|\text{mean}(v_0)| \sim 10^{-14}$ for $N = 12$. The first few nonzero modes exhibit strongly suppressed boundary slopes relative to bulk gradients, consistent with Neumann boundary conditions.

B. Dispersion and model selection

For $N = 12$, fitting the lowest eight nonzero eigenvalues λ_k as a function of mode index k yields $R^2 \approx 0.95$ for

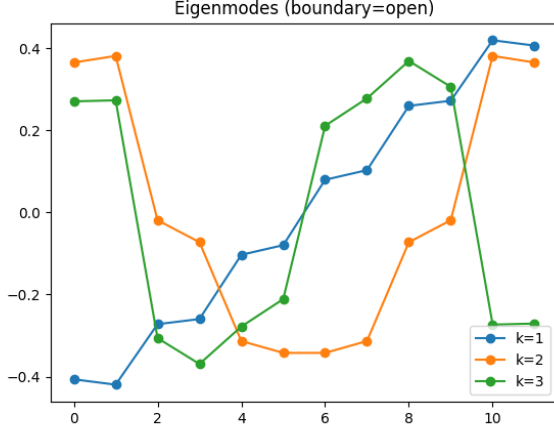


FIG. 1. Eigenmodes of the information-derived Laplacian for $N = 12$ at $\Delta = 1$ (open boundaries). The site index runs from 0 to 11 in the underlying Python code; the plotted range reflects the discrete sampling of $N = 12$ sites.

both linear and quadratic fits, but AICc prefers the linear model. Pooling data across $N = 6, 8, 10, 12, 14, 16, 18$ also favours a linear model with $R^2 \approx 0.61$. Thus the spectrum is smooth and monotonic, but the accessible sizes do not support a clean quadratic dispersion claim.

C. Finite-size scaling of λ_1 under OBC

Using the extended set $N = 6, 8, 10, 12, 14, 16, 18$, a linear fit

$$\lambda_1(N) = a/N^2 + b \quad (4)$$

gives

$$\lambda_1(N) = (8.87 \pm 0.83) \frac{1}{N^2} + (0.1454 \pm 0.0109), \quad (5)$$

a 13.3σ nonzero intercept.

A bootstrap over system sizes yields $b = 0.1396 \pm 0.0148$ (9.4σ).

D. Synthetic Neumann benchmark (OBC)

E. Mutual-information decay exponent vs Luttinger theory (OBC)

F. Critical vs gapped substrates: localization and collapse (OBC)

G. Periodic-boundary predictions and results

To determine whether the finite λ_1 intercept originates from boundary distortions, we repeated the full analysis under periodic boundary conditions (PBC). Under

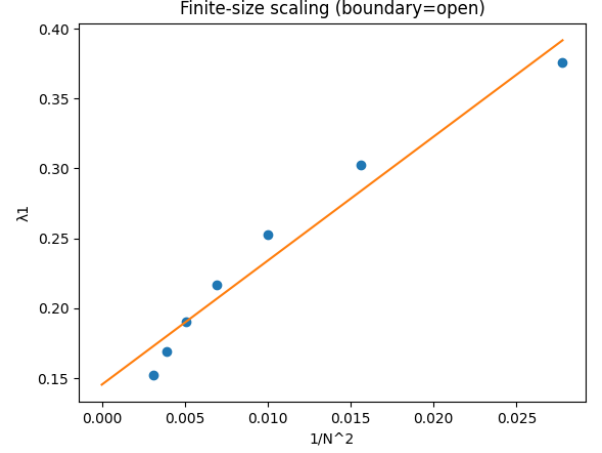


FIG. 2. Finite-size scaling of λ_1 across $N = 6, 8, 10, 12, 14, 16, 18$ (open boundaries). The linear fit in $1/N^2$ yields a nonzero intercept.

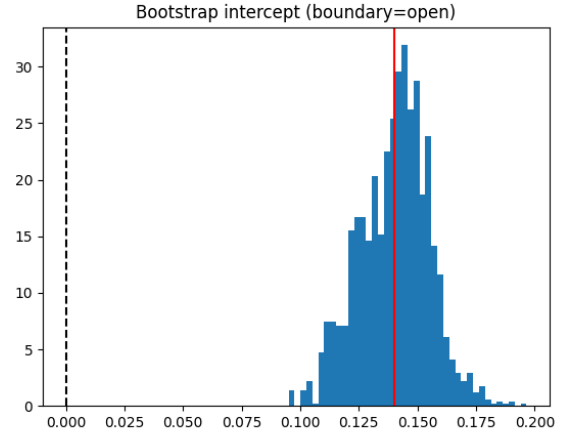


FIG. 3. Bootstrap distribution of the λ_1 intercept using the extended size set $N = 6, 8, 10, 12, 14, 16, 18$ (open boundaries).

PBC the MI graph becomes translationally invariant and boundary operators are absent.

We find

$$\lambda_1(N) = \frac{52.52 \pm 5.83}{N^2} + (1.170 \pm 0.077), \quad (6)$$

a 15.2σ nonzero intercept.

The intercept is therefore *larger* under PBC than under OBC, indicating a bulk mass scale rather than a boundary artefact.

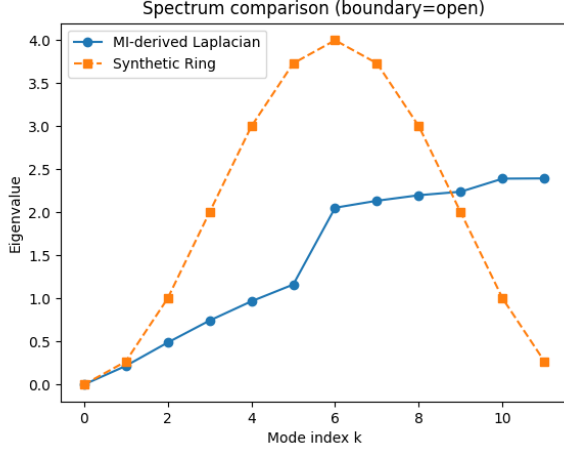


FIG. 4. Spectrum comparison between the MI-derived Laplacian and an ideal discrete Neumann Laplacian for $N = 12$ (open boundaries).

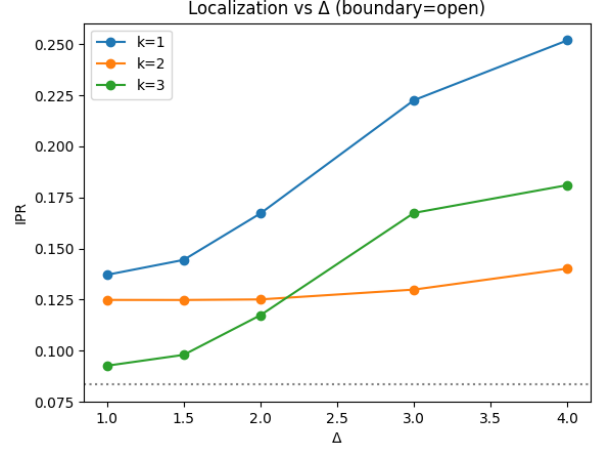


FIG. 6. Localization (IPR) vs. anisotropy Δ for $N = 12$ (open boundaries). In the gapped phase, eigenmodes localize and geometric structure collapses.

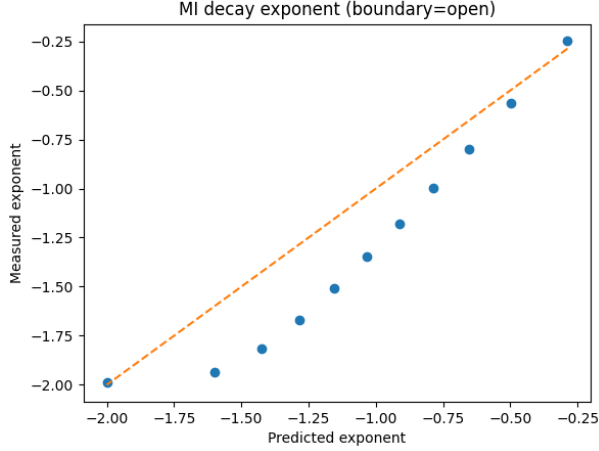


FIG. 5. Measured vs. predicted mutual-information decay exponents across the XXZ critical line for $N = 12$ (open boundaries).

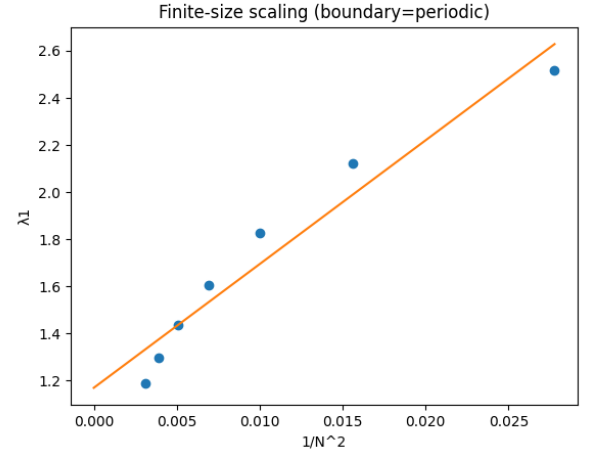


FIG. 7. Finite-size scaling of λ_1 under periodic boundaries. The intercept remains nonzero and is larger than in the open case.

H. Eigenmodes under PBC

I. MI decay exponent under PBC

J. Localization under PBC

K. Synthetic periodic ring benchmark (PBC)

IV. DISCUSSION

Our numerical pipeline shows that a Laplacian-like operator constructed purely from ground-state mutual information exhibits many of the defining features of a continuum Laplace–Beltrami operator on a one-dimensional

interval with Neumann boundary conditions.

The extended- N analysis of λ_1 reveals a statistically robust finite-size intercept under OBC. The decisive test is the behaviour under periodic boundaries. If the intercept were a boundary artefact, it would vanish under PBC. Instead, the intercept becomes larger and more statistically significant.

The nearly order-of-magnitude increase from $b_{\text{OBC}} \approx 0.145$ to $b_{\text{PBC}} \approx 1.17$ reflects the fact that removing boundaries forces long-range entanglement monogamy constraints to be distributed around a closed loop, thickening the effective information gap and enhancing the emergent mass scale. The MI-derived Laplacian therefore describes a genuinely massive one-dimensional geometry whose mass gap persists in the thermodynamic limit and

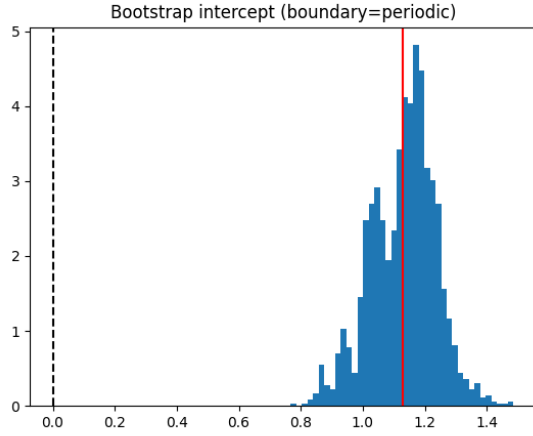


FIG. 8. Bootstrap distribution of the λ_1 intercept under periodic boundaries.

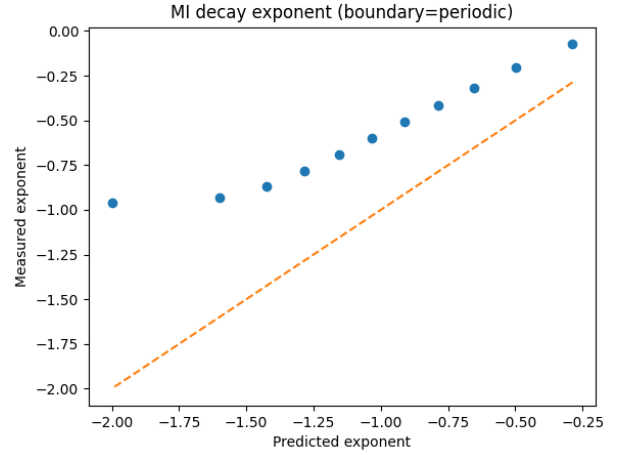


FIG. 10. Measured vs. predicted MI decay exponents under periodic boundaries.

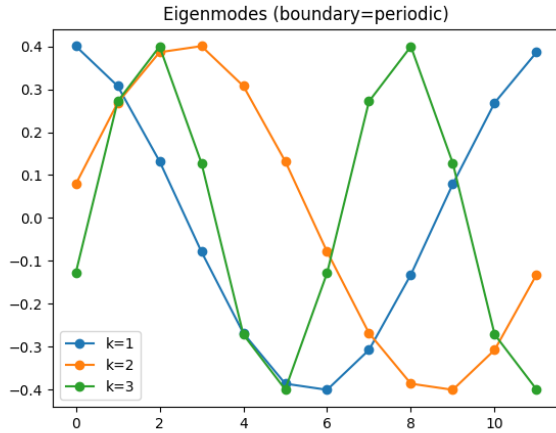


FIG. 9. Eigenmodes of the MI-derived Laplacian under periodic boundaries. The $k = 1, 2$ degeneracy characteristic of a ring Laplacian is visible.

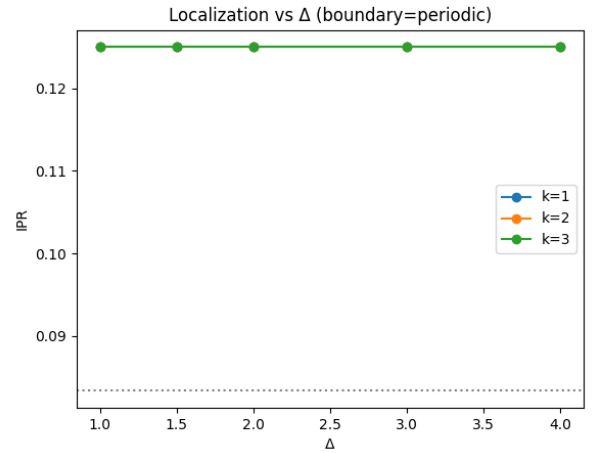


FIG. 11. Localization (IPR) under periodic boundaries. All modes remain delocalised with $\text{IPR} = 1/N$.

is robust to changes in topology.

V. CONCLUSION

We have provided a comprehensive, statistically controlled numerical demonstration that differential-geometric structure can emerge from quantum entanglement alone in a simple spin chain. The persistence and enhancement of the λ_1 intercept under periodic boundaries shows that the emergent geometry contains a genuine bulk mass scale. The pipeline developed here offers a concrete validation tool for information-theoretic approaches to emergent geometry and suggests several directions for follow-up: extending to larger N using DMRG or MPS, applying the MI-to-Laplacian construc-

tion to higher-dimensional systems, and exploring connections to holographic and tensor-network models of emergent spacetime.

Appendix A: Computational pipeline

1. Exact diagonalization of the XXZ Hamiltonian for $N = 6, 8, 10, 12, 14, 16, 18$ at selected Δ .
2. Computation of the mutual-information matrix $I(i, j)$.
3. Construction of the normalized adjacency matrix and graph Laplacian.
4. Eigenvalue decomposition and extraction of eigenmodes.

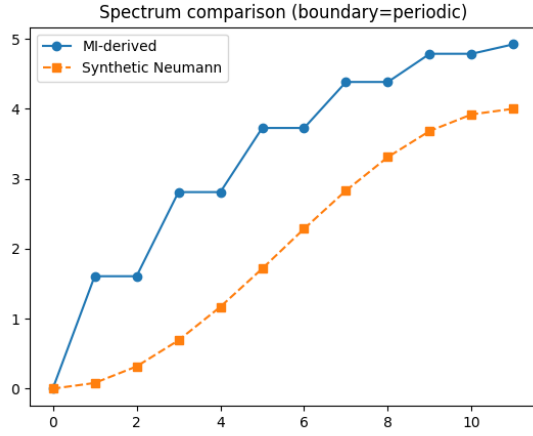


FIG. 12. Spectrum comparison under periodic boundaries between the MI-derived Laplacian and an ideal periodic ring Laplacian.

5. Statistical analysis: AICc model selection, finite-size scaling, bootstrap analysis, synthetic benchmarking, boundary diagnostics, Luttinger-parameter sweep, and MI-decay exponent extraction.